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Anomaly Detection

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Hitesh Mohapatra

KIIT University

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Intrusion Detection Systems

Lecture #8

Intrusion Detection Techniques

Anomaly Detection

Anomaly Detection

- What are Anomalies?
- Principles of Anomaly Detection: Classification Based Anomaly Detection, Anomaly Detection Model
- Advantages & Limitations of Anomaly Detection
- Challenges in Anomaly Detection
- Output of Anomaly Detection
- Classification of Anomaly Detection Systems
 - Statistical Based Methods
 - Machine Learning Based Methods
 - Data Mining Based Methods
- Case Study

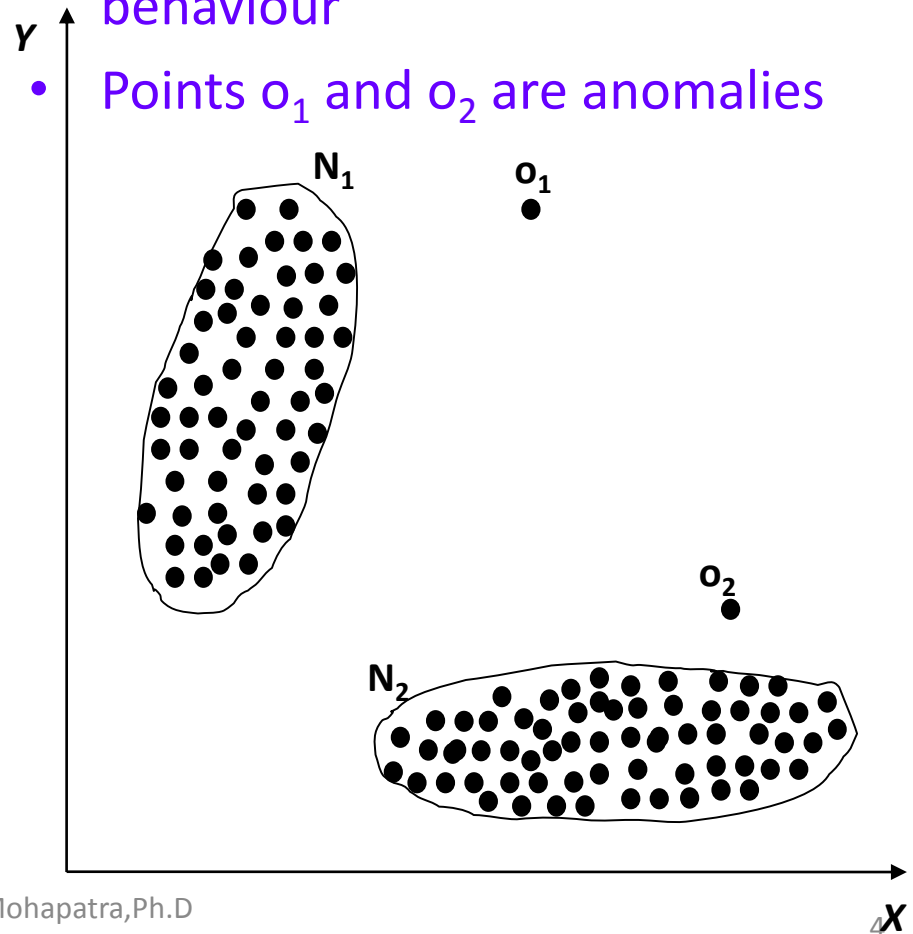
What are Anomalies?

- **Anomaly** is a pattern in the data that does not confirm to the expected behaviour
- Anomalies occur very infrequently among the enormous amount of data/transactions that is generated in many scientific, commercial and other real life applications
- Also referred to as **outliers**, exceptions, etc.
- Detection of anomalies has recently gained a lot of attention in many security domains, ranging from intrusion detection to detection of fraudulent transactions in real life applications
- **Real Life Anomalies:** Credit card fraud, Insurance claim fraud, Mobile/Cell phone fraud, etc.

Anomalies as Outliers

- Although anomalies or outliers occur very infrequently, their impact is quite high compared to other events, making their detection extremely important

- N_1 and N_2 are regions of normal behaviour
- Points o_1 and o_2 are anomalies



Anomaly Detection

- Assumption
 - Intrusions are necessarily abnormal in terms of user behavior or system behavior
- Focus on
 - Finding and defining what is normal user/system/network behavior
- Build some model to represent the “normal behavior”
 - Alert is raised if the current user/system behavior deviates substantially from normal behavior

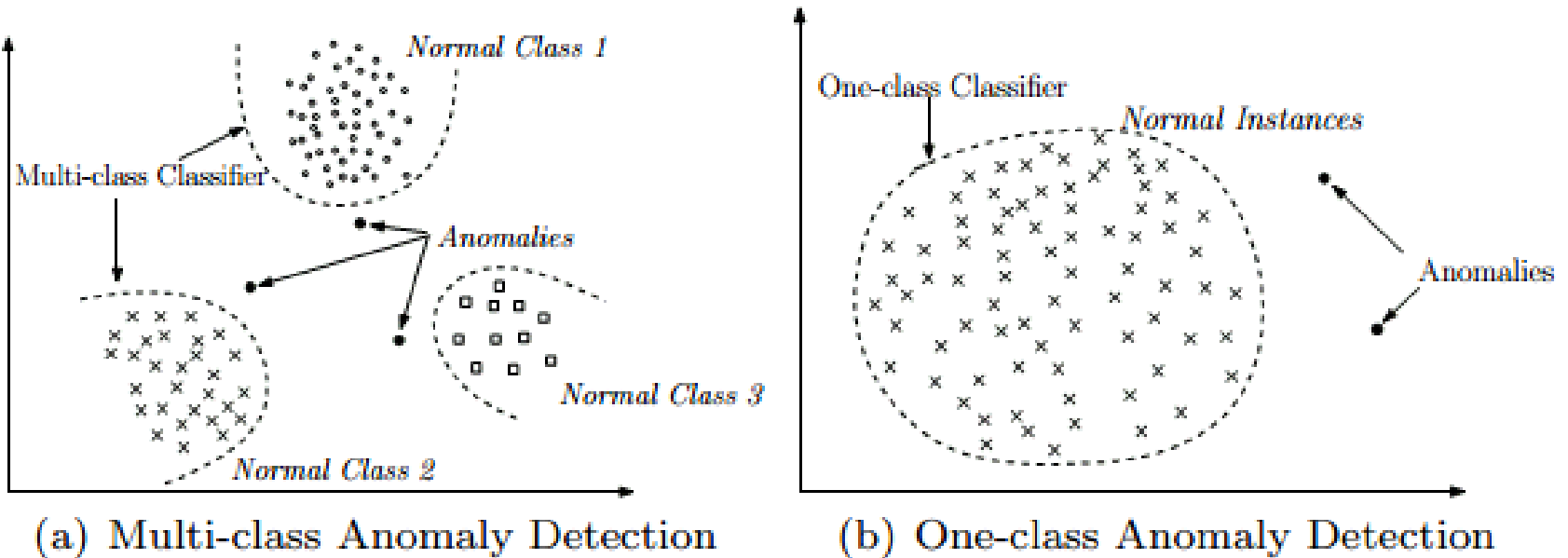
Classification Based Anomaly Detection

- **Classification** is used to train a model (classifier) from a set of labeled data instances (training) and then, classify a test instance into one of the classes (normal or anomalous) using the learnt model (testing)
- An anomaly detection approach roughly consists of two phases:
 - **Training Phase:**
 - An anomaly detection system first creates a profile of the normal system, network, or program activity
 - **Testing Phase:**
 - Learned profile is applied to new data (Any activity that deviates from the profile is treated as a possible intrusion)
 - A test instance is classified as normal or anomalous using the classifier

Classification Based Anomaly Detection

- One-class classification based anomaly detection techniques
 - Assume that all training instances have only one class label
 - Any test instance that does not fall within the learnt boundary is declared as anomalous
- Multi-class classification based anomaly detection techniques
 - Assume that the training data contains labeled instances belonging to multiple normal classes
 - A test instance is considered anomalous if its not classified as normal by any of the classifiers

Classification Based Anomaly Detection



Anomaly Detection Model

- A typical anomaly detection model is illustrated in Figure 1:

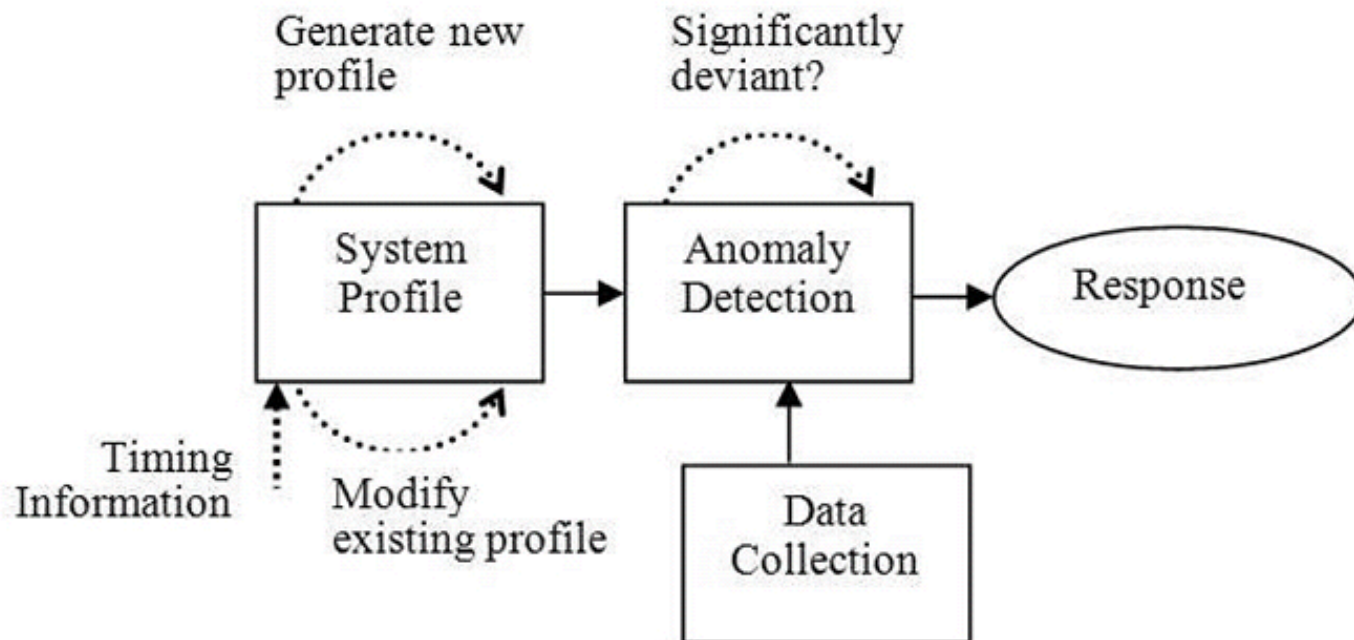


Figure 1: Anomaly Detection Model

Anomaly Detection Model

- It consists of four components – data collection, system profile, anomaly detection and response
- Normal user activities or traffic data are obtained and saved by the data collection component
- Specific modeling techniques are used to create “normal profiles”
- The anomaly detection component decides how far the current activities deviate from the normal system profiles and accordingly what percentage of these activities should be flagged as abnormal
- Finally, the response component reports the intrusion and sometimes corresponding timing information

Advantages of Anomaly Detection

- Anomaly detection systems offer several benefits:
 - The primary advantage of anomaly detection is has the **potential to detect novel attacks/previously unknown attacks**
 - *Anomaly detection systems addresses the biggest limitation of misuse detection systems*
 - A second advantage of anomaly detection systems is that:
 - *Profiles of normal activity are customized for every system, application and network*
 - *This makes very difficult for an attacker to know with certainty what activities it can carry out without getting detected*
 - Anomaly detection systems have the capability to detect **zero-day attacks** as well as **insider attacks**

Disadvantages of Anomaly Detection

- Due to the underlying assumptions of anomaly detection mechanisms, their **false alarm rates** are in general very high compared to misuse detection systems
- The main reasons for this limitation include the following:
 - The user's normal behavior model is based on data collected over a period of time
 - The effectiveness of anomaly detection is heavily dependent on **how accurately the normal behavior is modeled and updated** over time
 - Any mistake in **choosing the parameters/features used for building the normal profile** will increase the false alarm rate and decrease the effectiveness of the anomaly detection system
- Usually more **computationally expensive** than misuse detection

Challenges in Anomaly Detection

- Accurate Representation of Normal Behavior of each user is very challenging
- Selection of the user, system or network features to be used for building the normal profile
- The boundary between normal and outlying behaviour is often not precise (exact)
- The exact notion of an outlier is different for different application domains
- Need To Be Adaptive to Accommodate Evolving User/System Behavior
 - *User behavior may evolve over time or there may be sudden change due to some requirements*
 - *System behavior could change due to upgrades of OS, library, compiler etc.*
- Availability of labelled data for training/validation

Output of Anomaly Detection

- The outputs produced by anomaly detection systems are one of the following two types:
 - **Scores:** Scoring techniques assign an anomaly score to each instance in the test data depending on the degree to which that instance is considered an anomaly
 - **Labels:** Techniques in this category assign a label (normal or anomalous) to each test instance

Anomaly Detection Techniques

- Many anomaly detection techniques have been proposed in the literature
- We can divide the Anomaly IDSs into the following categories according to the technique involved in the “behavioral model” considered :
 - Statistical Based Methods
 - Machine Learning Based Methods
 - Data Mining Based Methods

Statistical Based Methods

- In statistical-based techniques, the user/system/network traffic activity is captured and a **normal profile** representing its behavior is created
- Statistical methods monitor the user/system/network behavior by **measuring certain variables over time** (E.g. login and logout time of each session)
- Two datasets are considered and compared during the anomaly detection process:
 - *Currently observed behavior*
 - *Previously trained statistical profile*

Statistical Based Methods

- As the system/network events are processed, the current activity is observed and an **anomaly score** estimated by comparison of the two behaviors – current and normal profile is computed.
- The score indicates the **degree of irregularity or intrusiveness** for a specific event.
- If the anomaly score is higher than a certain **threshold**, the IDS generates an alert.

Statistical Based Methods

- **Univariate Models:**

- **Univariate** refers to an expression, equation, function or polynomial of only one variable
- The earliest statistical approaches, both network oriented and host oriented IDS, corresponded to univariate models
- Parameters are modeled as independent variables thus defining an acceptable range of values for every variable (**Denning and Neumann, 1985**)

- **Multivariate Models**

- This model consider the correlations between two or more variable/metrics (**multivariate**)
- These are useful because experimental data have shown that a better level of discrimination can be obtained from combinations of related measures rather than individually (**Ye et al., 2002**)

Statistical Based Methods

- Time Series Models:
 - This model uses **time-related metrics** (interval timer, together with an event counter or resource measure/usage over a period of time)
 - It takes into account the **order and the inter-arrival times** of the observations as well as their values
 - An observed behavior will be labeled as abnormal if it deviates significantly from the normal patterns established using the time-related metrics

Statistical Based Methods

- Advantages:
 - Statistical methods provide accurate notification of malicious activities
 - These models have the ability to learn the expected behavior of the monitored system from observations
 - Such systems have the capability of detecting “zero day” or the very latest attacks
- Disadvantages:
 - Skilled attackers can train a statistical anomaly detection to accept abnormal behavior as normal
 - It can also be difficult to determine thresholds that balance the likelihood of false positives and false negatives
 - All behaviors cannot be modeled using statistical methods

Case Study: Haystack

- Haystack [1] is one of the earliest host-based statistical anomaly IDS
- It uses both user and group-based anomaly detection strategies by maintaining a database of user groups and individual profiles
- Models system parameters/features as independent, Gaussian random variables
- It defines acceptable behavior for a user within a particular user group

Case Study: Haystack

- A set of features such as the amount of I/O, CPU utilization, number of file accesses are observed
- Haystack defined a range of values that are considered normal for each feature
- If during a session, the value of some feature falls outside the normal range, the score for the subject is raised and this is reported as an intrusion
- If a user had not previously been detected, a new user profile was created using restrictions based on the user's group membership

Drawbacks of Haystack

- One drawback of Haystack was that it was designed to work offline. The attempt to use it for real-time intrusion detection systems failed, since doing so required high-performance systems
- Secondly, because of its dependence on maintaining profiles, a common problem for system administrators was the determination of what attributes were good indicators of intrusive activity

Case Study: IDES Statistical Anomaly Detector

- SRI IDES Statistical Anomaly Detector
 - Published in *IEEE Symposium on Security and Privacy*, 1991
 - Developed at the Stanford Research Institute (SRI) and was called the *Intrusion Detection Expert System (IDES)* [2, 3]

IDES Statistical Anomaly Detector

- The SRI IDES system is a **real-time intrusion detection expert system** that observes behavior on a monitored computer system
- It adaptively learns what is normal for individual users, groups, remote hosts and the overall system
- Observed behavior is flagged as a potential intrusion if it deviates significantly from expected behavior or it triggers a rule in the expert system rule base
- The model is based on a **multivariate statistical engine**

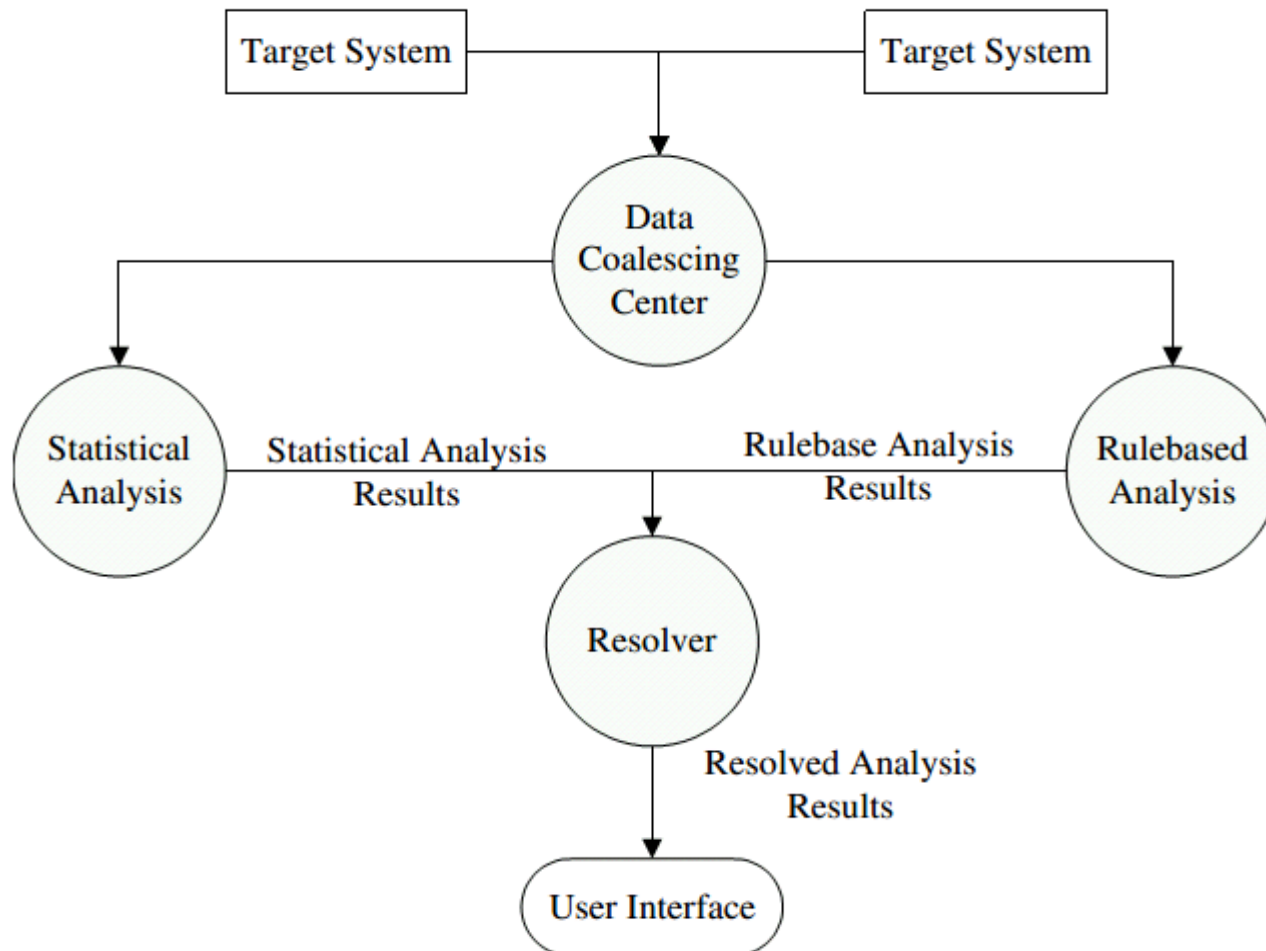
Case Study: Next-Generation Intrusion Detection Expert System (NIDES)

- Afterwards, an improved version of IDES called the **Next-Generation Intrusion Detection Expert System (NIDES)** was proposed in 1995, which is a hybrid system [4, 5]
- NIDES is a centralized, multihost-based hybrid detection (anomaly and misuse) system that performs real-time monitoring of user activity
- Audit data are collected from the multiple target hosts and provided to the two analysis components:
 - **Statistical analysis component** (Anomaly-based)
 - **Rule based analysis component** (Misuse-based)

Next-Generation Intrusion Detection Expert System (NIDES)

- The **audit data** collected consist of:
 - User names, names of files accessed, total number of files opened, number of pages read from secondary storage, identities of machines onto which user has logged, etc.
 - NIDES stores only statistics related to frequencies, means, variances, etc. of measures instead of the total audit data

Flow Chart of Real Time Operation in NIDES



Statistical Analysis Component

- A **subject** is a user of a computer system
- The statistical approach used in NIDES compares a subject's **short-term behavior** with the subject's **historical** or **long-term behavior**
- Short-term behavior is more concentrated on specific activities and long-term behavior is distributed across many activities
- The **NIDES statistical component** compares short-term and long-term behaviors to determine whether they are statistically similar and keeps track of the amount of deviation between the two behaviors

NIDES Measures

- Aspects of **subject behavior/profile** are represented as **measures** (e.g., names of files accessed, CPU usage, hour of use, etc.)
- For each measure, **probability distribution** is constructed for short-term and long-term behaviors
- For example, for the measure of file access (probabilities are attached to the file names):
 - **Long-term probability distribution** would consist of the historical probabilities with which different files have been accessed
 - **Short-term probability distribution** would consist of the recent probabilities with which different files have been accessed

NIDES Measures

- In the case of continuous measures, such as CPU usage time, the probabilities are attached to a ranges of values
- The collection of measures and their long-term probability distributions as defined as the **subject's profile**
- The NIDES measures are classified into four groups:
 - Activity Intensity
 - Audit Record Distribution
 - Categorical
 - Continuous

NIDES Measures

- The **activity intensity measure** determines whether the volume of activity generated is normal
- The **audit record distribution measure** determines whether, for recently observed activity (say, the last few hundred audit records generated), the types of actions being generated are normal
- The **categorical and continuous measures** determine whether, within a type of activity (say, file access or CPU usage time), the types of actions carried out are normal

Half-life

- The number of audit records or days of audit record activity that constitute short-term and long-term behavior can be set through the specification of a **half-life**
- **Half-life** is the number of audit records that need to be refreshed before the contribution of a given data item is decayed (down weighted) by one half
- For the long-term probability distributions, the half-life is set at 30 days
- With this setting, audit records that were gathered 30 days back contribute $\frac{1}{2}$ as much weight as the recent records, 60 days past audit records contribute $\frac{1}{4}$ th weight and so on...

Aging Rate

- The **aging rate** is a multiplicative factor less than or equal to unity, by which the existing information in a profile is aged
- The smaller the rate, the more rapidly this information is “forgotten”
- For example, if the aging rate is 0.8, the third most recent audit record has a weight of $0.8 \times 0.8 \times 0.8$ or 0.512

Rule Based Analysis Component

- NIDES comes with a rule base analysis component
 - Rules are encoded in the rule base relating to:
 - Known attacks and intrusion scenarios
 - Specific actions or patterns of behavior that are suspicious or known security violations
 - Expert system looks for matches between current activity and rules in the rule base
- Rule base can also be extended and updated in NIDES

Next-Generation Intrusion Detection Expert System (NIDES)

- Combining the values obtained for each measure and taking into consideration the correlation between measures, the IDS computes an index of how far the current audit record is from the normal state
- An anomaly is flagged if the audited activity is sufficiently far from the expected behavior (beyond a threshold)
- Adaptive historical profiles for each “user” are maintained
 - Updated regularly
 - Old data “aged” out during profile updates

Next-Generation Intrusion Detection Expert System (NIDES)

- **Resolver** in NIDES produce alerts that aims at:
 - Removing false alarms
 - Removing false negatives
 - Direct notification to the appropriate authority

Machine Learning Based Methods

- Machine learning can be defined as the ability of a program or a system to learn and improve its performance over time
- Machine learning techniques are based on establishing a **model/classifier** that enables the current patterns to be categorized as normal or malicious
- Learning mechanism incorporates learning capabilities into the intrusion detection process
- Machine learning techniques focus on building a IDS that has the ability to change its execution strategy (for improving performance) based on the newly acquired information

Machine Learning Based Methods

- Data Labels

- The labels associated with a data instance denote if that instance is normal or anomalous
- Labeling is often done manually by a human expert and hence requires substantial effort, time and cost to obtain the labeled training data set
- Getting a labeled set of anomalous data instances which cover all possible type of anomalous behavior is more difficult and challenging than getting labeled data for normal behavior
- The anomalous behavior is often dynamic and sometimes unpredictable in nature
- **E.g., Air traffic safety** – New types of anomalies might arise, for which there is no labeled training data resulting in devastating consequences

Machine Learning Based Methods

- Based on the extent to which the labeled data are available, anomaly detection techniques can operate in one of the following three modes:
 - Supervised
 - Semi-Supervised
 - Unsupervised

Machine Learning Based Methods

- Supervised Anomaly Detection
 - Establishes the normal profiles of systems/networks through training based on labeled data sets
 - **Assumption:** Availability of training data sets which has labeled instances for normal as well as anomaly classes
 - The main drawbacks of supervised anomaly detection are:
 - Need of labeled training data, which makes the process error-prone, costly and time consuming and difficult to find new attacks
 - The no. of anomalous instances in the labeled training data are much less compared to the normal instances

Machine Learning Based Methods

- Semi-Supervised anomaly detection
 - **Assumption:** Techniques that operate in a semi-supervised mode assume that the training data has labeled instances for only the normal class/classes
 - As they do not require labels for the anomaly class, they are more widely applicable than supervised techniques
 - The typical approach used in such techniques is to build a model for the normal behavior and use the model to identify anomalies in the test data
 - For example, in space craft fault detection an anomaly scenario would signify an accident, which is not easy to model

Machine Learning Based Methods

- Unsupervised Anomaly Detection
 - Techniques that operate in unsupervised mode do not require training data (attacks or normal instances), and thus are most widely applicable
 - The techniques in this category make the assumption that *normal instances are far more frequent than anomalies in the test data*
 - If this assumption is not true then such techniques suffer from high false alarm rate

Machine Learning Techniques

- The following Machine learning Based techniques are popularly used for modeling anomaly detection systems:
 - Neural Networks
 - System Call based Sequence Analysis
 - Bayesian Networks
 - Markov Models

Machine Learning Based Methods

- Neural Networks

- With the aim of simulating the operation of the human brain, neural networks have been adopted in the field of anomaly intrusion detection
- This detection approach has been employed to create user profiles, to build a prediction model, to identify the intrusive behavior of traffic patterns, etc.

Machine Learning Based Methods

- Neural Networks

- A basic anomaly detection technique using neural networks operates in two steps:

- First, a neural network is trained on the normal training data to learn the normal class/classes
 - Second, each test instance is provided as an input to the neural network to test whether it is normal or anomalous

Case Study: Intrusion Detection with Neural Networks

- This paper proposes a new way of applying neural networks to detect intrusions - **Neural Network Intrusion Detector (NNID)** [6]
- NNID is a **backpropagation neural network** trained to identify users based on identifying a legitimate user based on the distribution of commands he/she executes
- The set of commands used and their frequency, constitutes a “print” (profile) of the user (possible to identify the user based on this information)

Case Study: Intrusion Detection with Neural Networks

- The NNID model is implemented in a UNIX environment
- The system administrator runs NNID at the end of each day to see if the user's sessions match with his normal patterns
- If a user's behavior does not match his/her profile, the system administrator is alerted of a possible security breach

Case Study: Intrusion Detection with Neural Networks

- NNID for a particular computer system consists of the following three phases:
 - 1) Collecting Training Data:
 - Audit logs are obtained for each user for a period of several days
 - For each day and for each user, NNID forms a vector that represents how often the user executed each command

Case Study: Intrusion Detection with Neural Networks

2) Training:

- The neural network is trained to identify the users based on their **command distribution vectors**

3) Performance/Testing:

- The network identifies a user by comparing his/her profile command distribution vector with each new command distribution vector
- If the network's output is different from the user's profile, it signals an anomaly

Machine Learning Based Methods

- System Call Based Sequence Analysis
 - Every program can be specified a set of system call sequences determined by the functions called in the program and their order in all possible execution paths
 - This method involves learning the behavior of a program and recognizing significant deviations from the normal
 - The overall idea is to *build up a separate database of normal behavior for each process of interest*
 - Once a stable database is constructed for a given program in a particular environment, the database can then be used to monitor the program's behavior

Machine Learning Based Methods

- System Call Based Sequence Analysis
 - The sequences of system calls in different programs forms the set of normal patterns/normal profile sequences
 - Programs that show system call sequences that deviates from the normal profile sequences are considered as symptoms of an attack/indicate anomalies
 - Forrest et al. [7] analyzes sequences of program's system calls in the UNIX operating system and uses them to build a normal profile for anomaly detection
 - They have analyzed several UNIX based programs and showed that sequences of system calls could be used to build a normal profile of a program

Machine Learning Based Methods

- System Call Based Sequence Analysis
 - There are two stages in the proposed algorithm [6]:
 - In the first stage, normal behavior traces are scanned to build up a database of normal patterns (observed sequences of system calls)
 - In the second stage, new traces are scanned that might contain abnormal behavior by looking for patterns not present in the normal profile

Machine Learning Based Methods

- Bayesian Network

- A Bayesian network is a graphical model that encodes probabilistic relationships among variables of interest
- It is a **Directed Acyclic Graph (DAG)** that represents a set of random variables and their conditional dependencies
- Can be used to model problems where there is a need to combine prior knowledge with observed data

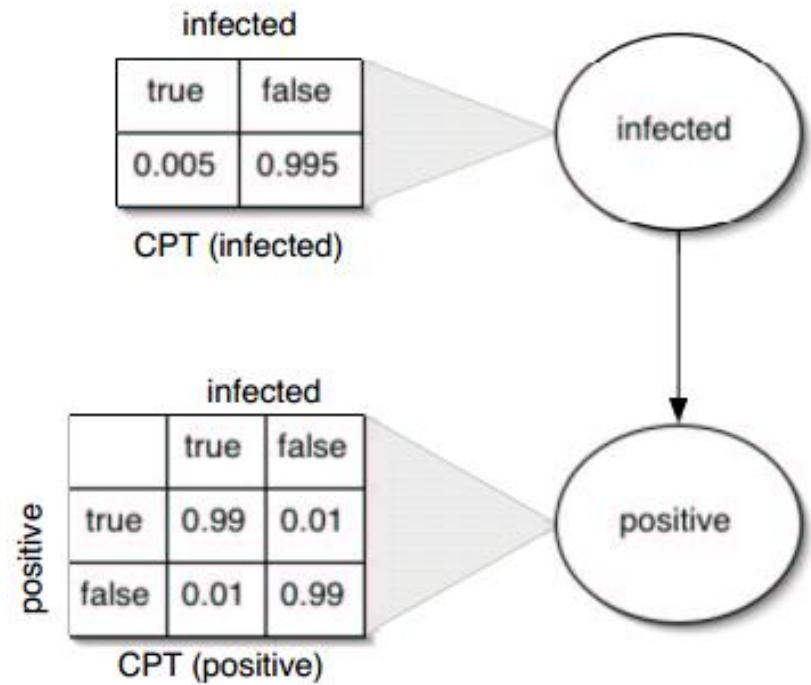
Machine Learning Based Methods

- Bayesian Network
 - Each node contains the states of the random variable that it represents and a Conditional Probability Table (CPT)
 - The CPT of a node contains probabilities of the node being in a specific state given the states of its parents
 - The purpose of a Bayesian network is to allow the calculation of the posterior probability of the hypothesis variable(s) given the support of the observed evidence (Bayesian Learning)

An Example of a Bayesian Network

- This situation can be represented with two random boolean variables, *infected* and *positive*
- The farmer has a test to determine whether the milk is infected or not
- The outcome of the test is either positive or negative
- The variable *infected* is true when the milk is actually infected and false otherwise
- The variable *positive* is true when the test claims that the milk is infected and false otherwise

- We consider the following example where a farmer has a bottle of milk that can be either infected or clean



Machine Learning Based Methods

- Bayesian Learning for Anomaly Detection:
 - Bayesian learning model encodes probabilistic relationships among variables of interest
 - Bayesian networks can be used for one-class and multi-class anomaly detection
 - Aggregates information from different variables and provide an estimate of the expectancy that event belong to one of normal or anomalous class

Machine Learning Based Methods

- Bayesian Learning for Anomaly Detection
 - Given a test data instance, IDS estimates the posterior probability of observing a class label (from a set of normal class labels and the anomaly class label)
 - The class label with largest posterior is chosen as the predicted class for the given test instance (MAP Hypothesis)
 - The likelihood of observing the test instance given a class (likelihood probabilities) and the prior probabilities are estimated from the training data set

Machine Learning Based Methods

- Posterior belief of test data is computed using Bayesian learning
- The prior/initial belief $P(h)$ can be updated by using **Bayes' Rule** after getting the new information D_i :

$$P(h|D_i) = \frac{P(D_i|h)P(h)}{P(D_i|h)P(h) + P(D_i|\bar{h})P(\bar{h})}$$

- The goal of Bayesian learning is to find the most probable hypothesis h_{map} given the training data (**Maximum A Posteriori Hypothesis**)

$$h_{\text{map}} = \max_{h' \in H} P(h'|D_i) \quad \text{where, } h' \in \{h, \bar{h}\}$$

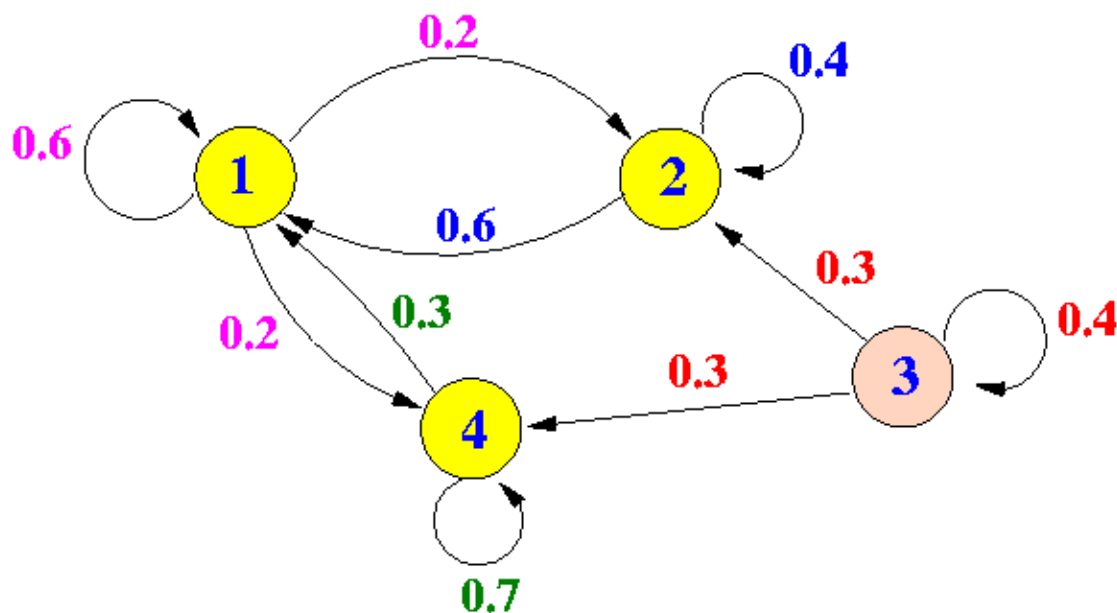
Machine Learning Based Methods

- Markov Models

- Markov chains, have also been employed extensively for anomaly detection
- A **Markov chain** is a set of states $S = \{s_1, s_2, \dots, s_n\}$ that are interconnected through certain transition probabilities
- The process starts in one of these states and moves successively from one state to another
- If the chain is currently in state s_i , then it moves to state s_j at the next step with a probability denoted by p_{ij} (transition probability)
- The matrix representing the transition probabilities of each state to all other states is known as the matrix of transition probabilities or the transition matrix

Machine Learning Based Methods

- An Example of a Markov Chain and Transition Probability Matrix



$$P = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix} \end{matrix}$$

Machine Learning Based Methods

- Markov Model for Anomaly Detection
 - Ye et al.[8], present an anomaly detection technique that is based on Markov chains
 - This paper presents a cyber-attack detection technique through anomaly detection and discusses the robustness of the modeling technique employed
 - In this technique, a Markov-chain model represents a profile of computer-event transitions in a normal/usual operating condition of a computer and network system (a normal profile)
 - The Markov-chain model of the normal profile is generated from historic data of the system's normal activities

Machine Learning Based Methods

- Markov Model for Anomaly Detection
 - The observed activities of the system are analyzed to infer the probability that the Markov-chain model of the norm profile supports the observed activities
 - The larger the probability, the more likely the sequence of states results from normal activities
 - A sequence of states from attack activities is assumed to receive a low probability of support from the Markov chain model of the normal profile

Data Mining Based Methods for Anomaly/Outlier Detection

- What are anomalies/outliers?
 - The set of data points that are considerably different than the remainder of the data
- Variants of Anomaly/Outlier Detection Problems
 - Given a database D , find all the data points $x \in D$ with anomaly scores greater than some threshold t
 - Given a database D , containing mostly normal (but unlabeled) data points, and a test point x , compute the anomaly score of x with respect to D
- Applications:
 - Credit card fraud detection, telecommunication fraud detection, network intrusion detection, fault detection

More Definitions of an Outlier

Hawkins' Definition of Outlier

An **outlier** is an observation that differs so much from other observation as to arouse suspicion that it was generated by a different mechanism

Moore and McCabe (1999)

An **outlier** is an observation that lies outside the overall pattern of a distribution

Data Mining Based Methods for Anomaly/Outlier Detection

- Assumption:
 - There are considerably more “normal” observations than “abnormal” observations (outliers/anomalies) in the data
- General Steps:
 - Build a profile of the “normal” behavior (Profile can be patterns or summary statistics for the overall population)
 - Use the “normal” profile to detect anomalies (Anomalies are observations whose characteristics differ significantly from the normal profile)

Approaches to Data Mining Based Anomaly Detection

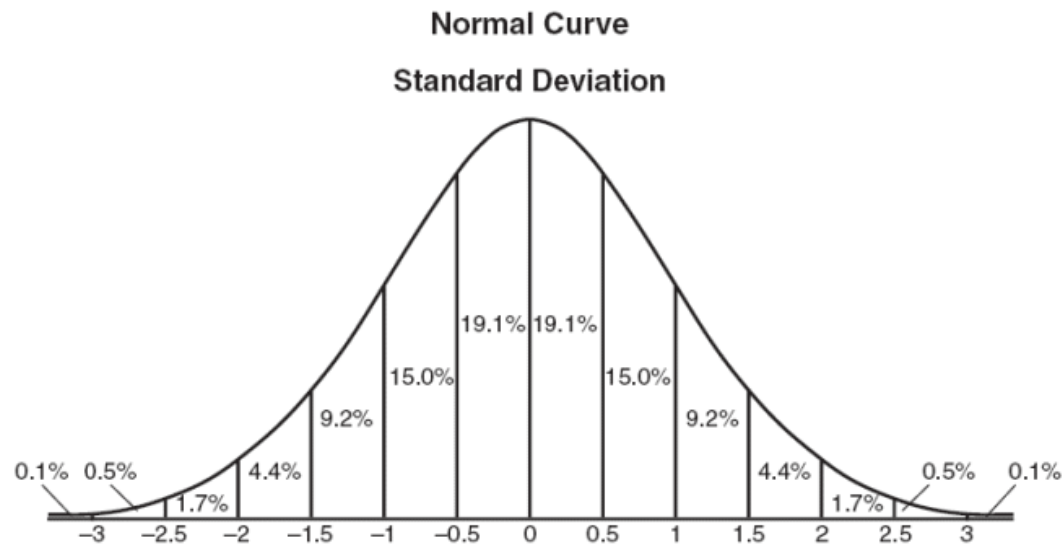
- Statistical Approaches
- Proximity-Based
- Density-Based

Statistical Approaches

- Statistical approaches were the earliest algorithms used for outlier detection
- Statistical Approaches are model-based
- A model is created for the data and objects are evaluated with respect to how well they fit into the model
 - This approach is based on building a **probability distribution model** that considers how likely objects belong to that model
- An **outlier** is an object that has a low probability with respect to a probability distribution model of the data (Probabilistic Definition of an Outlier)

Statistical Approaches

- This approach assumes a model describing the distribution of the data (e.g., normal distribution)
- Statistical models are generally suited to quantitative real-valued data sets which are suitable for statistical processing



Statistical Approaches

- Issues of Statistical Approaches:
 - No training data is available
 - Specific Distribution: The choice of statistical distribution for the data
 - Number of attributes: single attribute or multivariate data
 - Number of distributions: modeling with a single one or a mixture of distributions
 - Detecting which objects are anomalous among huge amount of data

Proximity-Based Technique

- Proximity-based techniques are simple to implement and make no prior assumptions about the data distribution model
- First a **proximity measure** is defined between the objects
- Anomalies are objects that are **distant** from most of the other objects
- The basic notion of this approach is:
 - An object is anomalous if it is distant from most points
- **Proximity measure:** often is chosen as distance so referred to as “**Distance-Based Outlier Detection Techniques**”

Proximity-Based Technique

- Distance Based Outlier Detection
 - Let N be the number of objects in the input dataset T and let DF be the underlying distance function that gives the distance between any pair of objects in T
 - An object O in a dataset T is considered to be a $DB(p,d)$ outlier if at least a fraction p of the objects in T lie at a distance greater than d from O
 - The clusters can be formed by using different attributes
 - The distance function DF can be computed by applying Euclidean distance

Proximity-Based Technique

- It is easier to determine a proximity measure for data set than to determine its statistical distribution
- One of the simplest way to measure whether an object is distant from most points (outlier) is to use the *k-nearest neighbor* approach
- An arbitrary instance is represented by $(a_1(x), a_2(x), a_3(x), \dots, a_n(x))$
where $a_i(x)$ denotes features
- The distance between two instances can be computed by applying *Euclidean distance*

Proximity-Based Technique

- Distance Based Outlier Detection
 - Euclidean Distance

$$d_E(\mathbf{x}, \mathbf{y}) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2} = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

– Example: $DF = \sqrt{(loc_diff)^2 + (time_diff)^2}$

where

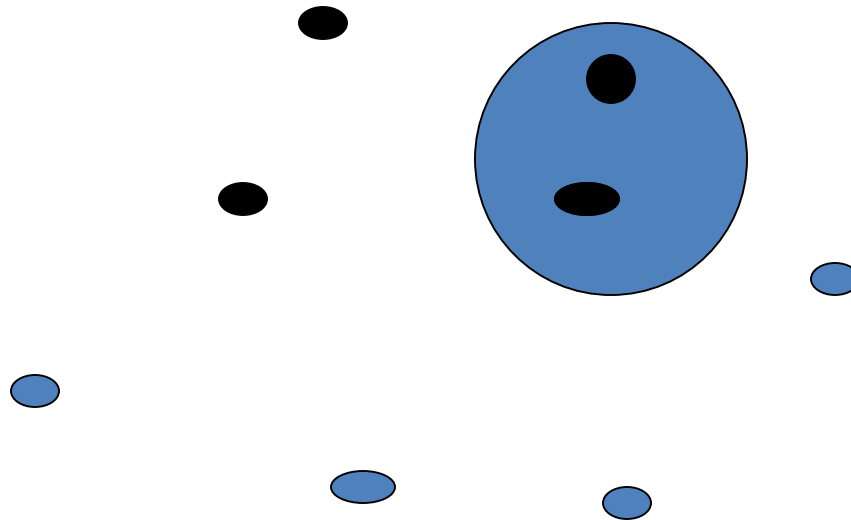
loc_diff: distance between current transaction location and the user's normal profile transaction location

time_diff: distance between current transaction time slot

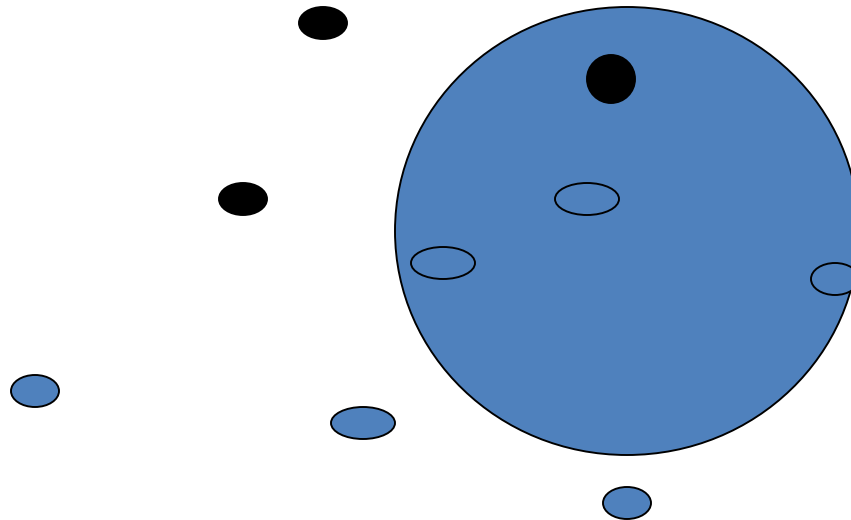
Proximity-Based Technique

- The **outlier score** of a data instance is defined as its distance to its k^{th} nearest neighbor in a given data set
- A threshold can be applied on the outlier score to determine if a test instance is an outlier or not

1-Nearest Neighbor



3-Nearest Neighbor



Density-Based Approaches

- For each point, compute the density of each data instance to compute its outlier score
- An object is anomalous if it's in a region of **low density**
- A degree of being an outlier is assigned to each object
- This degree is called the **Local Outlier Factor (LOF)** of an object which signifies its degree of outlierness
- The degree depends on how isolated the object is with respect to the surrounding neighborhood

Density-Based Approaches

- The LOF of an object is based on the single parameter of MinPts, which is the number of nearest neighbors used in defining the local neighborhood of the object
- DBSCAN (Density Based Spatial Clustering of Applications with Noise) is a density based clustering algorithm [9] which can be used to filter out outliers and discover clusters of arbitrary shapes

Density-Based Approaches

- The key idea of the DBSCAN algorithm is that for each point p in a cluster c_i , there are at least a minimum number of points (MinPts) in the neighborhood of that point p
- The density in the neighborhood of each point p has to exceed some threshold
- If MinPts is set to 1, then each point in the database is treated as a separate cluster
- The higher the value of MinPts, less is the number of clusters formed

Density-Based Approaches

- A transaction is detected as an outlier if it does not belong to any cluster in the cluster set
- Such an observation gives evidence that the transaction could be fraudulent
- The extent of deviation of an incoming transaction is measured by its degree of outlierness

References

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